

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Paper No. BPT-503: Quantum Mechanics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. Symbols have their usual meanings.***Question 1.** Answer any **five** of the following:

[5 × 4 = 20]

- (a) Is it possible to perform the Compton effect experiment using visible light? Explain.
- (b) What is the length of a 1D infinite square well box in which an electron in the ground ($n = 1$) state has the same energy as a photon of wavelength 600 nm?
- (c) Show that for a relativistic particle of rest energy E_0 , the de Broglie wavelength is given by:

$$\lambda = \frac{hc}{E_0 \sqrt{\gamma^2 - 1}} = \frac{hc}{E_0 \beta \gamma}$$

where $\beta = v/c$ and $\gamma = 1/\sqrt{1 - \beta^2}$.

- (d) Atoms are in two excited states with energies E_1 and E_2 such that $\Delta E = |E_1 - E_2|$. These atoms return to the ground state by emitting photons. Show that the difference in the emitted wavelengths is:

$$\Delta\lambda \approx \frac{\lambda^2}{hc} \Delta E$$

- (e) Using the uncertainty principle, estimate the ground state energy of a one-dimensional linear harmonic oscillator described by $H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$.
- (f) If $\langle T \rangle$ and $\langle V \rangle$ are the expectation values of the kinetic and potential energy operators for the wave function $\psi(x)$, find the new expectation values of these operators if the wave function undergoes scaling to $\psi(\alpha x)$, where α is a real constant.
- (g) The normalized wave functions ψ_1 and ψ_2 correspond to the ground state and first excited state of a particle in a potential. If the operator A acts such that $A\psi_1 = \psi_2$ and $A\psi_2 = \psi_1$, find the expectation value of the operator A for the state $\Psi = \frac{3\psi_1 + 4\psi_2}{5}$.

Question 2. For a particle of mass m in a one-dimensional infinite potential box of length L (extending from $x = 0$ to $x = L$):

- (a) Find the wave functions and energy eigenvalues of the ground and the first excited states.
[6.5]
- (b) Sketch the wave functions and probability densities for the ground and the first excited states.
[5.5]

Question 3.

- (a) Normalize the wave function $\psi(x, t) = Ne^{-\lambda|x|}e^{-iEt/\hbar}$ and determine the expectation values of x and x^2 . [6]
- (b) Show that the energy eigenvalue E must be greater than the minimum value of the potential $V(x)$ for any normalizable solution of the time-independent Schrödinger equation. [4]
- (c) Show that Planck's constant h has the same unit as angular momentum. [4]

OR

- (a) The ground state wave function of an electron in a hydrogen atom is:

$$\psi_0(r) = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}$$

where a_0 is the Bohr radius.

- (i) Show that the wave function is normalized. [4]
- (ii) Find the probability of finding the electron between $r_1 = a_0/2$ and $r_2 = 3a_0/2$. [5]
- (iii) Determine the uncertainty in the radial position of the electron in the ground state. Use the expectation value $\langle r \rangle = 3a_0/2$ and $\langle r^2 \rangle = 3a_0^2$. [5]

Question 4.

- (a) The first excited state of a particle in a linear harmonic oscillator is given by:

$$\psi_1(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \sqrt{\frac{2m\omega}{\hbar}} x e^{-m\omega x^2/2\hbar}$$

Calculate the ground state wave function by applying the lowering operator a_- . [9]
(Hint: $a_- = \frac{1}{\sqrt{2m\hbar\omega}}(ip + m\omega x)$)

- (b) Let $\vec{L}$ and $\vec{p}$ be the angular and linear momentum operators, respectively, for a particle. Evaluate the commutation relations $[L_z, p_y]$ and $[x^2, p_x^2]$. [5]

Question 5.

- (a) A particle of mass m is contained in a one-dimensional infinite well extending from $x = -L/2$ to $x = L/2$. The particle is in its ground state given by $\psi_0(x) = \sqrt{\frac{2}{L}} \cos\left(\frac{\pi x}{L}\right)$. The walls of the box are moved suddenly to form a box extending from $x = -L$ to $x = L$. What is the probability that the particle will be in the ground state of the new box after this sudden expansion? [7]
- (b) Show that the angular momentum operators L^2 and L_z commute. Prove that these operators share the same eigenstates. [7]

OR

- (a) Write down the Schrödinger equation for a linear harmonic oscillator and solve it using power series method to obtain the eigenvalues and eigenfunctions. [14]